



Internship Project — Final

Project Title:

Real-Time Fraud Detection System with Explainable AI & Live Dashboard

Assigned Date: 13/05/2026

Submission Date: 25/05/2026

Level: Advanced

Domain: AI & Data Analytics

Week 4 — Capstone



Problem Statement

Financial fraud costs the global economy over \$5 trillion annually. Credit card fraud, identity theft, and transaction anomalies demand systems that can detect suspicious activity in milliseconds — not after the damage is done. Legacy rule-based systems miss novel attack patterns; simple ML models are black boxes that compliance teams can't trust.

Your task is to build an end-to-end Fraud Detection System that combines state-of-the-art machine learning, severe class imbalance handling, and Explainable AI (SHAP) to produce a system that a real fraud analyst could deploy and trust.

You will go beyond accuracy metrics — you will justify every prediction to a non-technical stakeholder and package results inside a live interactive dashboard.

This project simulates the work of a Machine Learning Engineer + Data Analyst at a fintech or banking firm. Treat it as your portfolio centerpiece.



Dataset — Where to Get It

Credit Card Fraud Detection — IEEE-CIS

Download from Kaggle:
<https://www.kaggle.com/c/ieee-fraud-detection/data>

Steps to download:

1. Create or log in to your free Kaggle account at www.kaggle.com
2. Search for “IEEE-CIS Fraud Detection”
3. Download:
 - train_transaction.csv
 - train_identity.csv
4. Use both files — they must be merged on TransactionID

Dataset Stats:

- 590,000 transactions
- 433 features
- 3.5% fraud rate

 Note: Use train_transaction.csv and train_identity.csv only. Do not use the test files.

Tasks to Complete

Complete all tasks inside a Jupyter Notebook (.ipynb).
Each task must be clearly labeled as a Markdown heading.

TASK 1 — Data Loading, Merging & Exploratory Analysis

- Load both CSVs and merge on TransactionID using Pandas
 - Display shape, dtypes, and first 10 rows of the merged dataset
 - Analyse the isFraud target column — quantify and visualize the class imbalance
 - Identify missing values column-by-column
 - Decide drop vs. impute threshold (suggest: drop columns with >50% missing)
 - Plot distribution of TransactionAmt for fraud vs. non-fraud (use log scale)
 - Compute a correlation heatmap of the top 20 numerical features using Seaborn
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TASK 2 — Preprocessing, Imbalance Handling & Feature Engineering

- Drop columns with more than 50% missing values
- Impute remaining values using:
 - Median (numerical)
 - Mode (categorical)
- Label-encode high-cardinality categorical columns
- Justify your encoding strategy in a Markdown cell

Create at least 3 engineered features:

Examples:

- $\text{AmtToMeanRatio} = \text{TransactionAmt} / \text{mean}(\text{TransactionAmt})$
- $\text{HourOfDay} = \text{extracted from TransactionDT}$
- $\text{DeviceRisk} = \text{binary flag based on DeviceType and DeviceInfo}$

Additional Tasks:

- Apply SMOTE only on the training set
 - Scale numerical features using RobustScaler
 - Perform stratified 80/20 train-test split
 - Report class ratio before and after SMOTE
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TASK 3 — Model Training, Comparison & Threshold Optimization

Train 3 classification models:

Models:

1. LightGBM Classifier
2. XGBoost Classifier
3. Isolation Forest

Evaluate Using:

- Accuracy
- Precision
- Recall

- F1-Score
- ROC-AUC
- PR-AUC

Visualizations:

- Confusion Matrix for each model
- ROC Curve
- Precision-Recall Curve

Advanced:

- Optimize threshold using Threshold vs F1-Score plot
- Tune best model using:
 - Optuna
 - OR RandomizedSearchCV

Document improvements in Markdown.

TASK 4 — Explainable AI with SHAP Values [ADVANCED]

- Install and run SHAP library
- Generate Global SHAP Summary Plot (top 20 features)
- Generate SHAP Waterfall Plots for:
 1. Confirmed fraud case
 2. Borderline case (~0.50 probability)
 3. Legitimate transaction

Additional:

- Generate SHAP Dependence Plot
 - Explain all 3 transactions in plain English
 - Compare SHAP importance vs model feature importance
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TASK 5 — Risk Segmentation & Fraud Pattern Analysis [ADVANCED]

Segment transactions using fraud probability:

Risk Tiers:

-  Critical Risk → probability ≥ 0.75
-  Suspicious → probability between 0.40 and 0.74
-  Clear → probability < 0.40

Perform:

- Count transactions in each tier
- Compute:
 - Average TransactionAmt
 - Device type distribution
 - Hour-of-day pattern

Create:

- Grouped bar chart comparing all tiers

Identify:

Top 3 fraud patterns from Critical Risk transactions.

TASK 6 — Streamlit Fraud Operations Dashboard [ADVANCED]

Build a multi-page Streamlit application.

Page 1 — Overview

Display:

- Total transactions
- Total fraud count
- Detection rate
- Average fraud amount

Page 2 — Transaction Explorer

- Searchable table
- Filterable transactions
- Live risk score by TransactionID

Page 3 — SHAP Explainer

- User enters TransactionID
- Display SHAP waterfall plot
- Show plain-English explanation

Additional Requirements:

- Sidebar filters
 - Plotly interactive charts
 - Deploy on Streamlit Community Cloud
 - Submit live URL in README.md
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TASK 7 — Visualizations (Minimum 5 Charts)

Required Charts:

1. SHAP Global Summary Plot
2. Fraud rate by hour of day
3. TransactionAmt distribution
4. Risk tier donut chart
5. Precision-Recall curve with optimal threshold

Bonus:

Interactive Plotly scatter plot:

- TransactionAmt vs HourOfDay
 - Colored by fraud probability
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TASK 8 — Insights & Business Recommendations

Write a structured section answering:

- Which model performed best and why?
- Why PR-AUC matters more than accuracy in fraud detection?
- Top 3 fraud signals identified by SHAP

- Common characteristics of Critical Risk transactions
 - 2 actionable fraud prevention policies
 - Estimated money saved annually
 - Model limitations
 - Additional data that could improve performance
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Tools & Libraries to Use

Tool	Purpose
Python 3.x	Main programming language
Jupyter / Colab	Run notebook
Pandas / NumPy	Data manipulation
LightGBM	Primary classifier
XGBoost	Comparison model
Scikit-learn	ML utilities
imbalanced-learn	SMOTE
SHAP	Explainable AI
Optuna	Hyperparameter tuning

Plotly Interactive visualizations

Streamlit Dashboard

Matplotlib / Seaborn Charts

What to Submit

Create folder:

FraudDetection_[YourName]

Inside include:

-  analysis.ipynb
 -  data/
 - train_transaction.csv
 - train_identity.csv
 -  dashboard/app.py
 -  dashboard/model.pkl
 -  model_comparison.png
 -  shap_summary.png
 -  charts/ folder
 -  summary.pdf or summary.docx
 -  requirements.txt
 -  README.md
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Submission Link

https://docs.google.com/forms/d/e/1FAIpQLSeQbkVrKn_UM6eZ6JoK9oxa_cXb1DbsDoH87ap_8MRu_RU5Sw/viewform?usp=publish-editor